

دانشگاه صنعتی خواجه نصیرالدین طوسی

Research Project · Digital image processing course

CARD: Correlation Aware Restoration with Diffusion

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1 Abstract

This report presents the paper *CARD: Correlation Aware Restoration with Diffusion* and the accompanying course implementation. CARD addresses a mismatch between common restoration assumptions and real camera noise: many methods assume independent Gaussian noise, while rolling-shutter sensors can produce spatially correlated noise. The paper solves this by whitening the measurement equation and then applying DDRM-style diffusion restoration in the whitened measurement space. The implementation in this repository follows the same problem and solution structure for denoising: it builds a patchwise covariance matrix, generates correlated noise, computes a whitening transform, and uses both a lightweight CPU baseline and a Colab diffusion notebook to demonstrate correlation-aware restoration.

2 Problem and Motivation

Classical and learning-based image restoration methods often start from the linear observation model

$$y = Hx_0 + z, \quad z \sim \mathcal{N}(0, \sigma_y^2 I),$$

where x_0 is the clean image, H is the degradation operator, and the noise is independent and identically distributed. This assumption makes DDRM efficient because the measurement update can be derived as independent one-dimensional Gaussian updates in the spectral basis of H .

Real sensors, especially rolling-shutter CMOS sensors, can violate this model. The CARD paper instead considers

$$y = Hx_0 + n, \quad n \sim \mathcal{N}(0, \sigma_y^2 \Sigma),$$

where Σ is a positive definite covariance matrix. The covariance couples nearby pixels and breaks the independence assumption used by standard DDRM. Ignoring this structure can leave row-wise artifacts, color noise, and over-smoothed reconstructions.

3 CARD Methodology

CARD is a training-free modification of DDRM. It requires two ingredients:

- an estimate or model of the correlated-noise covariance Σ ;
- a frozen pretrained diffusion model used as an image prior.

3.1 Step 1: Whitening

CARD first preconditions the measurement equation with $W = \Sigma^{-1/2}$:

$$Wy = WHx_0 + Wn.$$

The whitened variables are

$$\tilde{y} = Wy, \quad \tilde{H} = WH, \quad \tilde{n} = Wn.$$

Since $W\Sigma W^T = I$, the transformed noise satisfies

$$\tilde{n} \sim \mathcal{N}(0, \sigma_y^2 I).$$

This restores the i.i.d. noise condition needed by DDRM, but for the modified operator $\tilde{H}$, not for the original operator H .

3.2 Step 2: Whitened DDRM Update

Standard DDRM uses the singular value decomposition

$$H = USV^T$$

and performs measurement conditioning in that spectral domain. CARD replaces this with the SVD of the whitened operator:

$$\tilde{H} = \tilde{U}\tilde{S}\tilde{V}^T.$$

The measurement coordinate becomes

$$\tilde{\tilde{y}} = \tilde{S}^\dagger \tilde{U}^T \tilde{y},$$

and the diffusion sample is projected as

$$\tilde{\tilde{x}}_t = \tilde{V}^T x_t.$$

The DDRM update then uses $\tilde{s}_i$, the singular values of $\tilde{H}$, rather than s_i . The effective measurement noise per spectral coordinate is

$$\delta_i = \frac{\sigma_y}{\tilde{s}_i}.$$

This is the core difference from ordinary DDRM: measurement consistency is computed after whitening, so correlated sensor noise is handled explicitly.

3.3 Patchwise Practical Approximation

The paper applies covariance and whitening over non-overlapping 8×8 patches per channel for efficiency. With patch dimension $d = 64$, CARD constructs a 64×64 covariance matrix, computes a coloring transform to synthesize correlated noise, and computes $W = \Sigma^{-1/2}$ for whitening. This same patchwise structure is used in the course implementation.

4 Paper Evaluation

The full paper evaluates CARD on denoising, deblurring, and super-resolution. It compares against restoration baselines such as DDRM and other methods under synthetic correlated noise and real correlated sensor noise. The paper also introduces CIN-D, a correlated image-noise dataset captured with a rolling shutter camera and dark frames for covariance estimation.

The expected output of a paper-level implementation is therefore not only a single denoised image. It is:

- restored images for denoising, deblurring, and super-resolution;
- comparisons against i.i.d.-noise baselines;
- PSNR, SSIM, and LPIPS metrics averaged over datasets;
- evidence that using Σ improves restoration under correlated noise.

5 Course Implementation

The repository contains two levels of implementation.

5.1 CPU Demonstration

The Python package `card` is a small reproducible demonstration of the image-processing part of CARD. It runs quickly on CPU and is useful for inspecting the covariance, coloring, whitening, and metric pipeline.

File	Responsibility
<code>card/noise.py</code>	covariance, coloring, whitening, noise generation
<code>card/restoration.py</code>	Gaussian baseline and covariance-aware denoising
<code>card/metrics.py</code>	PSNR and global SSIM metrics
<code>card/images.py</code>	synthetic image and output grid generation
<code>card/pipeline.py</code>	end-to-end CPU experiment
<code>main.py</code>	command-line entry point

Table 1: Main files in the CPU demonstration.

The CPU version does not reproduce DDRM. It replaces the diffusion prior with a simple Gaussian image prior and Wiener-style covariance-domain correction. This keeps the concept inspectable, but it is not the full CARD algorithm.

5.2 Colab Diffusion Notebook

The notebook `card_guided_diffusion_colab.ipynb` is closer to the paper. It uses a frozen OpenAI guided-diffusion checkpoint, patchwise covariance whitening, and an SVD-based measurement update for denoising where $H = I$. In that case,

$$\tilde{H} = WH = W.$$

The notebook runs multiple deterministic test images, applies correlated Gaussian noise, restores them with the CARD-style whitened diffusion update, and plots PSNR/SSIM comparisons against the noisy inputs and a Gaussian baseline.

6 How to Run

The CPU demo uses `uv`:

```
uv sync
uv run python main.py
```

The demo writes:

```
results/card_demo.png
```

Verification commands:

```
uv run pytest
uv run ruff check .
uv run basedpyright
```

The diffusion notebook should be run in Google Colab with a GPU runtime because it downloads OpenAI guided-diffusion and a pretrained checkpoint.

7 Implementation Results

The CPU demonstration output is shown in Figure 1. It is a small deterministic denoising example with a clean synthetic image, correlated measurement, Gaussian baseline, and covariance-aware restoration.

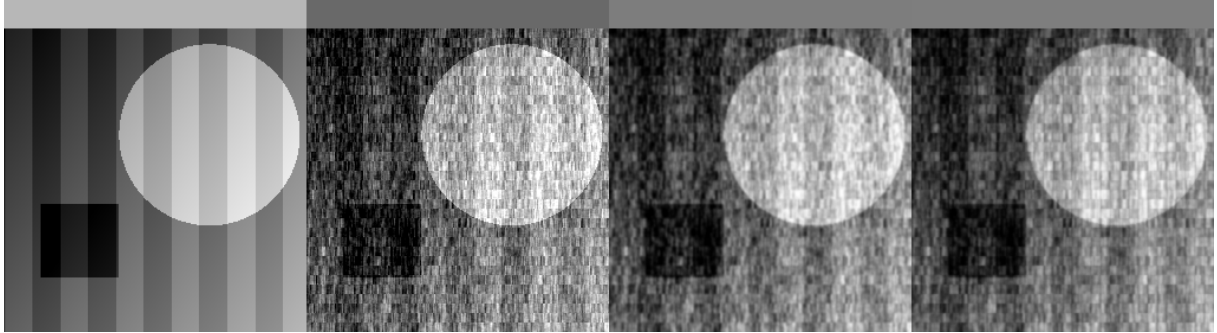


Figure 1: CPU demonstration output. Left to right: clean image, correlated noisy image, Gaussian baseline, and covariance-aware restoration.

Image	PSNR	SSIM
Correlated noisy	18.62	0.889
Gaussian baseline	22.07	0.946
Covariance-aware restore	22.39	0.949

Table 2: Measured quality in the deterministic CPU demo.

The Colab notebook is the preferred implementation artifact for presenting the paper methodology because it includes a pretrained diffusion prior and the whitened spectral measurement update. Its results should be interpreted as a course-scale denoising demonstration, not as the benchmark table from the paper.

8 Alignment and Limitations

The main limitation is scope. The current implementation demonstrates the central CARD idea for denoising, but it does not yet implement the full DDRM codebase, real dark-frame covariance estimation, CIN-D evaluation, deblurring, super-resolution, or LPIPS.

Component	CARD paper	This project
Noise model	Correlated Gaussian $n \sim \mathcal{N}(0, \sigma_y^2 \Sigma)$	Implemented
Whitening	$W = \Sigma^{-1/2}$, $W \Sigma W^T = I$	Implemented
Diffusion prior	Frozen pretrained diffusion model	Implemented in Colab
DDRM update	Closed-form update in SVD of $\tilde{H}$	Approximated in Colab
Tasks	Denoising, deblurring, super-resolution	Denoising only
Data	ImageNet, LSUN, CIN-D	Synthetic test images
Metrics	PSNR, SSIM, LPIPS	PSNR and global SSIM

Table 3: Alignment between the paper and the course implementation.

9 Conclusion

CARD shows that restoration methods should not treat correlated camera noise as i.i.d. noise. By whitening the measurement equation and running diffusion restoration in the SVD basis of the whitened operator, CARD preserves DDRM’s training-free structure while making the measurement model more realistic. The course implementation follows this logic at demonstration scale: it makes the covariance explicit, verifies whitening, generates correlated noise, and uses a Colab diffusion notebook to restore images under the denoising case $H = I$.